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# =====  
# Image Aliasing and Quantization Demonstration  
# Google Colab  
# =====  
  
import cv2  
import numpy as np  
import matplotlib.pyplot as plt  
from google.colab import files  
  
# Upload image  
uploaded = files.upload()  
  
image_name = list(uploaded.keys())[0]  
  
# Read image  
img = cv2.imread(image_name)  
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)  
  
# -----  
# 1. Original Image  
# -----  
  
# -----  
# 2. Aliasing (Low Sampling)  
# Downsample heavily and enlarge again  
# -----  
  
small = cv2.resize(img, (img.shape[1]//8, img.shape[0]//8),  
                    interpolation=cv2.INTER_NEAREST)  
  
aliased = cv2.resize(small,  
                     (img.shape[1], img.shape[0]),  
                     interpolation=cv2.INTER_NEAREST)  
  
# -----  
# 3. Quantization  
# Reduce image to only 16 intensity levels  
# -----  
  
levels = 16  
  
quantized = np.floor(img / (256/levels)) * (256/levels)  
quantized = quantized.astype(np.uint8)  
  
# -----  
# 4. Corrective Approach  
# Anti-aliasing using Gaussian Blur before Downsampling  
# -----
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blurred = cv2.GaussianBlur(img, (5,5), 0)

small_corrected = cv2.resize(
    blurred,
    (img.shape[1]//8, img.shape[0]//8),
    interpolation=cv2.INTER_AREA
)

corrected = cv2.resize(
    small_corrected,
    (img.shape[1], img.shape[0]),
    interpolation=cv2.INTER_CUBIC
)

# -----
# Display Results
# -----

plt.figure(figsize=(15,10))

plt.subplot(2,2,1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")

plt.subplot(2,2,2)
plt.imshow(aliased)
plt.title("Aliasing (Low Sampling)")
plt.axis("off")

plt.subplot(2,2,3)
plt.imshow(quantized)
plt.title("Quantization (16 Levels)")
plt.axis("off")

plt.subplot(2,2,4)
plt.imshow(corrected)
plt.title("Corrected (Anti-Aliasing)")
plt.axis("off")

plt.tight_layout()
plt.show()

# -----
# Observation
# -----

print("\nCAUSE OF ALIASING")
print("- Image was sampled at a very low resolution.")
print("- High-frequency details are lost, producing jagged edges.")
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print("\nQUANTIZATION EFFECT")
print("- Pixel intensity values are reduced to only 16 levels.")
print("- This causes visible banding and loss of smooth gradients.")

print("\nCORRECTIVE APPROACH")
print("- Apply Gaussian Blur before downsampling.")
print("- Use INTER_AREA for shrinking.")
print("- Use INTER_CUBIC when enlarging.")
print("- This reduces aliasing artifacts.")
```

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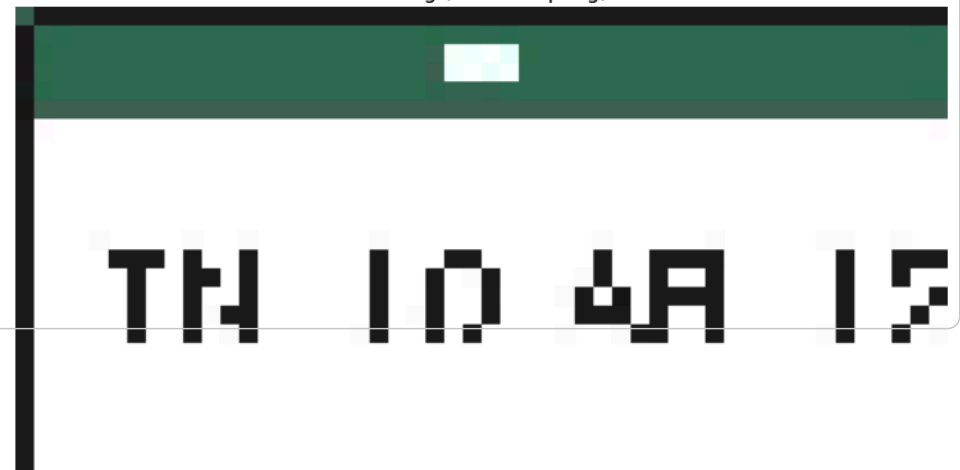
Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving number_plate.jpg to number_plate.jpg

Original Image



Aliasing (Low Sampling)



Quantization (16 Levels)



Corrected (Anti-Aliasing)

